

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	5 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	User Interface	USN-1	Design a responsive home page with project information and navigation	2	High	Member-1
Sprint-1	Credit Card Prediction	USN-2	As a user, I can enter applicant details such as income, employment status, and credit history	3	High	Member-2
Sprint-1	Credit Card Prediction	USN-3	As a user, I can receive an instant credit card approval or rejection prediction	5	High	Member-2
Sprint-2	Data Preprocessing	USN-4	Clean, preprocess, encode, and transform applicant data for model training	3	High	Member-3
Sprint-2	Machine Learning Model	USN-5	Train and compare Logistic Regression, Decision Tree, Random Forest, and XGBoost models	5	High	Member-2
Sprint-2	Backend Integration	USN-6	Integrate the trained machine learning model with the Flask web application	3	High	Member-1
Sprint-3	Prediction Dashboard	USN-7	Display prediction results along with applicant details in a user-friendly interface	3	Medium	Member-1
Sprint-3	Model Evaluation	USN-8	Evaluate model performance using accuracy, confusion matrix, and classification report	3	Medium	Member-3
Sprint-3	Testing	USN-9	Perform functional testing, unit testing, and validate prediction accuracy	2	High	Member-4
Sprint-4	Deployment	USN-10	Deploy the application using IBM Watson Machine Learning and IBM Cloud	3	Medium	Member-4
Sprint-4	Documentation	USN-11	Prepare the project report, user manual, GitHub	2	High	Member-4

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			repository, and presentation			

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	13	7 Days	05 June 2026	11 June 2026	13	11 June 2026
Sprint-2	11	7 Days	12 June 2026	18 June 2026	11	18 June 2026
Sprint-3	8	7 Days	19 June 2026	25 June 2026	8	25 June 2026
Sprint-4	5	7 Days	26 June 2026	02 July 2026	5	02 July 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

Total Story Points = 37

Sprint Duration = 28 Days

Average Velocity (Story Points per Sprint)

$$\text{Velocity} = \frac{37}{4} = 9.25 \approx 9 \text{ Story Points/Sprint}$$

Average Velocity (Story Points per Day)

$$\text{Velocity} = \frac{37}{28} = 1.32 \approx 1.3 \text{ Story Points/Day}$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

